



ISEL
INSTITUTO SUPERIOR DE
ENGENHARIA DE LISBOA

DE10-Lite Exp. Board

v1.0

Reference Manual

2022



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Overview

This document describes the hardware features of the DE10-Lite Expansion Board, including the detailed pin-out and component reference information required to create custom digital designs based on DE10-Lite Development Board.



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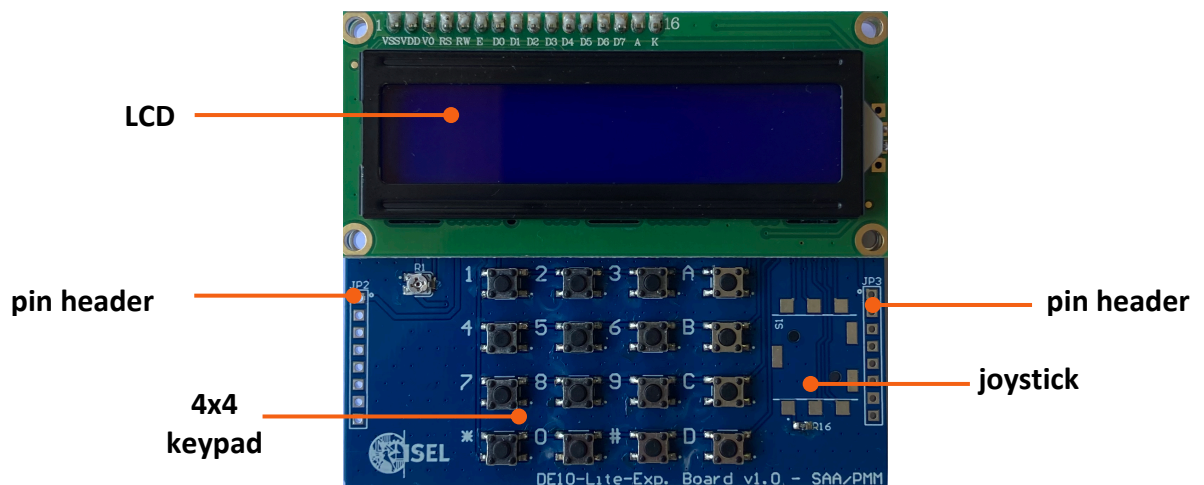


1 INTRODUCTION

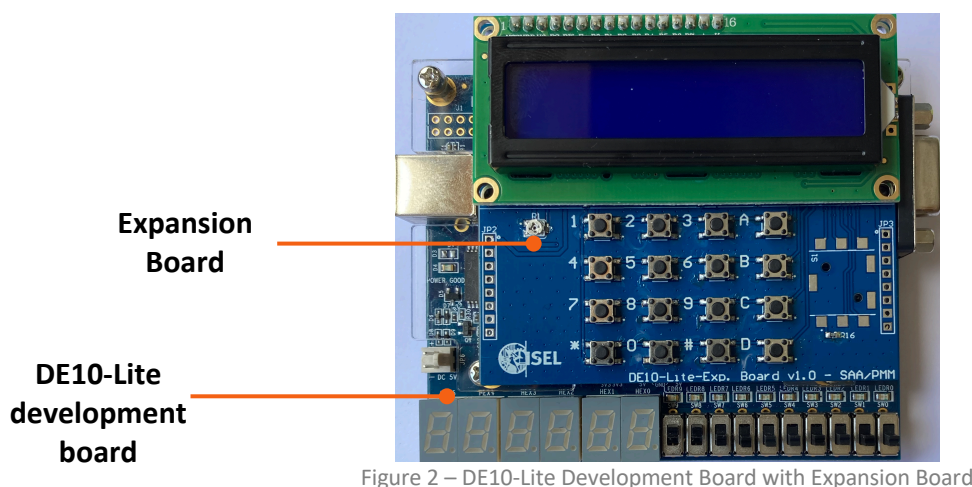
The DE10-Lite-Exp. Board is an expansion board for the DE10 lite Development Board that includes a Liquid Crystal Display (LCD), a 4x4 matrix keypad, one placeholder for a joystick (for future expansion) and two placeholders for pin headers (for future expansion).

For more information on the DE10-Lite Development Board, refer to the [DE10-Lite User Manual](#).

1.1 Board Layout



1.2 DE10-Lite with expansion board





2 USING DE10-LITE EXPANSION BOARD

2.1 Liquid Crystal Display (LCD)

The Liquid Crystal Display (LCD) is a two (2) lines and sixteen (16) columns text char matrix, with a four (4) or eight (8) bit interface. In this development board only write operations to the LCD are allowed. The LCD uses a HD44780U Dot Matrix Liquid Crystal Display Controller.

For more information on the HD44780U Dot Matrix Liquid Crystal Display Controller, refer to the [HD44780U Datasheet](#).

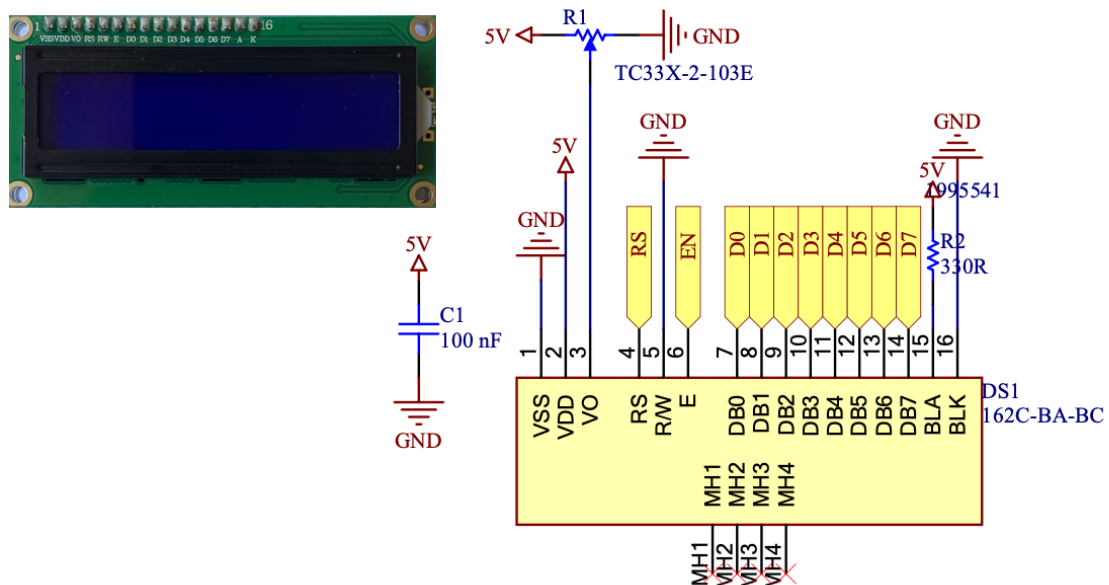


Figure 3 - Liquid Crystal Display (LCD) schematic

Table 1 – Pin Assignment of Liquid Crystal Display (LCD)

Signal Name	FPGA Pin No.	Description	I/O Standard
LCD_RS	PIN_W8	RS	3.3-V LVTTL
LCD_EN	PIN_V5	Enable	3.3-V LVTTL
LCD_DATA[0]	PIN_AA15	D0	3.3-V LVTTL
LCD_DATA[1]	PIN_W13	D1	3.3-V LVTTL
LCD_DATA[2]	PIN_AB13	D2	3.3-V LVTTL
LCD_DATA[3]	PIN_Y11	D3	3.3-V LVTTL
LCD_DATA[4]	PIN_W11	D4	3.3-V LVTTL
LCD_DATA[5]	PIN_AA10	D5	3.3-V LVTTL
LCD_DATA[6]	PIN_Y8	D6	3.3-V LVTTL
LCD_DATA[7]	PIN_Y7	D7	3.3-V LVTTL



2.2 4x4 Keypad

The 4x4 Keypad is organized in a four by four matrix, the four lines have pull-up resistors, while the columns have protection resistors.

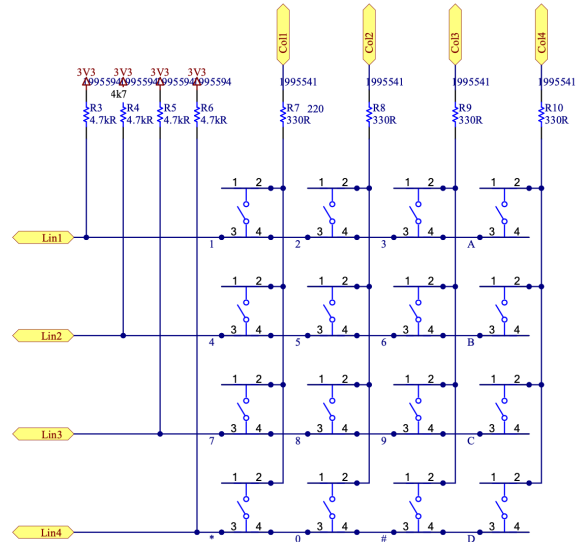
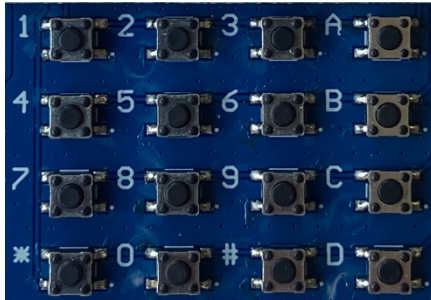


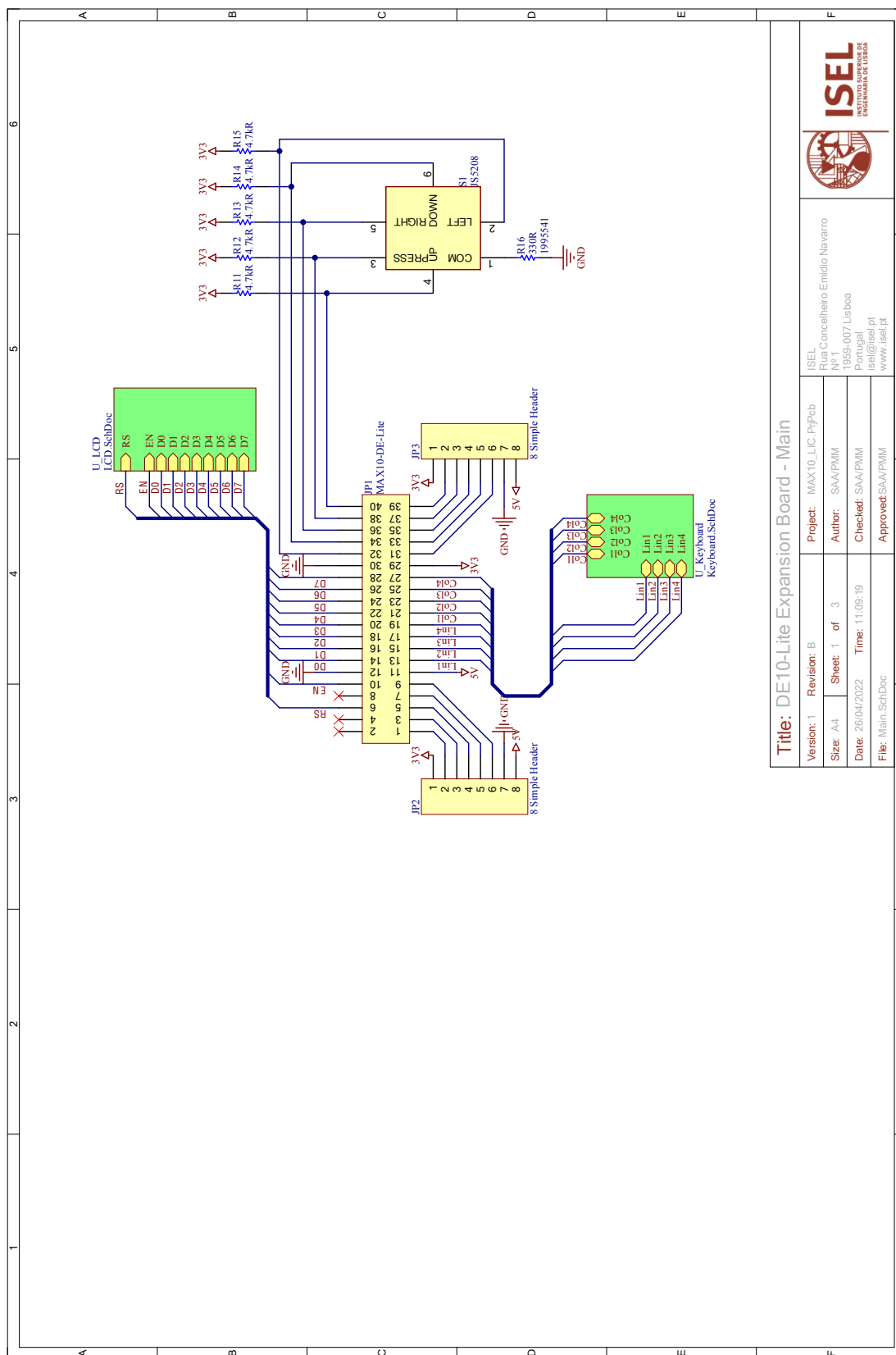
Figure 4 – 4x4 Keypad schematic

Table 2 – Pin Assignment of 4x4 Keypad

Signal Name	FPGA Pin No.	Description	I/O Standard
KEYPAD_LIN[0]	PIN_W5	4x4 keypad line 1	3.3-V LVTTL
KEYPAD_LIN[1]	PIN_AA14	4x4 keypad line 2	3.3-V LVTTL
KEYPAD_LIN[2]	PIN_W12	4x4 keypad line 3	3.3-V LVTTL
KEYPAD_LIN[3]	PIN_AB12	4x4 keypad line 4	3.3-V LVTTL
KEYPAD_COL[0]	PIN_AB11	4x4 keypad column 1	3.3-V LVTTL
KEYPAD_COL[1]	PIN_AB10	4x4 keypad column 2	3.3-V LVTTL
KEYPAD_COL[2]	PIN_AA9	4x4 keypad column 3	3.3-V LVTTL
KEYPAD_COL[3]	PIN_AA8	4x4 keypad column 4	3.3-V LVTTL

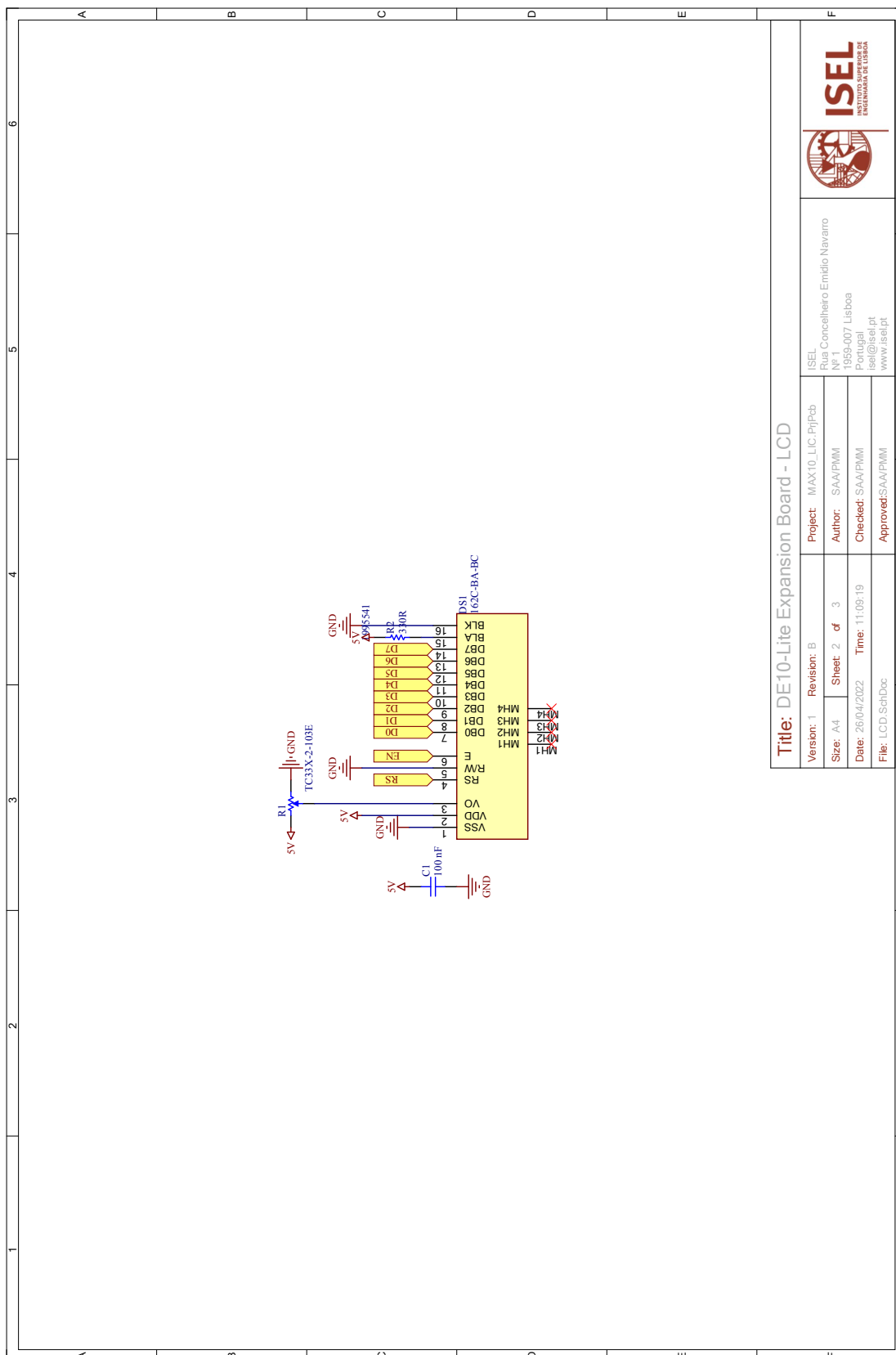


3 BOARD SCHEMATICS



Title: DE10-Lite Expansion Board - Main

Version: 1			Revision: B		ISEL Rua Concelheiro Emídio Navarro Nº1 1959-007 Lisboa Portugal isel@isel.pt www.isel.pt  ISEL INSTITUTO DE ENGENHARIA DE LISBOA
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